Chapter 5 – Roots of Equations

To solve $f(x)=ax^2+bx+c=0$

we use the quadratic formula $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

Roots are also called as zeros of the function.

* $f(x)=e^{-x}-x$ can not be solved analytically.

Math Background:

Algebraic Equations: A function $y=f(x)$ is algebraic if it can be expressed as:

$$f_n y^n+f_{n-1}y^{n-1}+\dots+f_1 y+f_0=0$$

where f_i : i^{th} order polynomial in x.

Transcendental Equations: non algebraic. Include trigonometric, exponential, logarithmic etc. functions.

EX $f(x)=\ln x^2-1$

$f(x)=e^{-0.2x}\sin(3x-0.5)$

* Roots of equations may be real or complex.

Preview of Finding Roots

1) Bracketing methods (start with guesses that contain the root)
   - Bisection
   - False position
   - Graphical methods
2) Open Methods (do not require that the initial guess bracket the root)(computationally efficient)

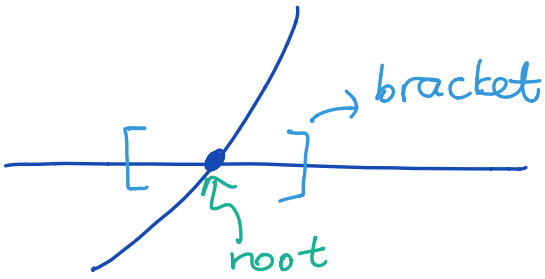
(do not always work)

- one-pt iteration
- Newton-Raphson
- Secant method

Bracketing Methods

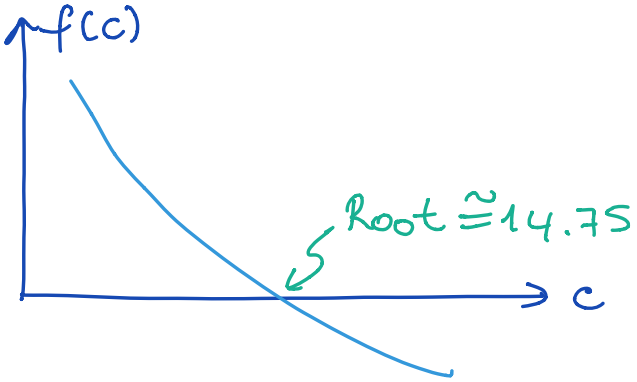
Fact: a function changes sign in the vicinity of a root.

* Bracketing methods require two initial guesses that bracket the root.

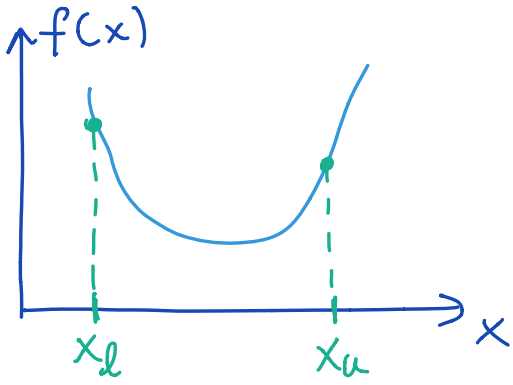


Graphical Methods: Make a plot of the function and observe where it crosses the x-axis.

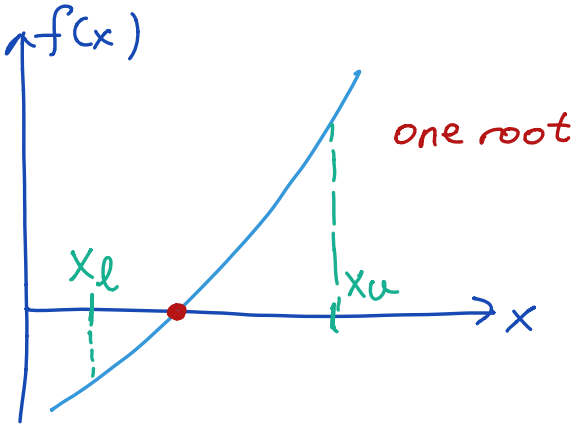
$$f(c) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right) - v$$

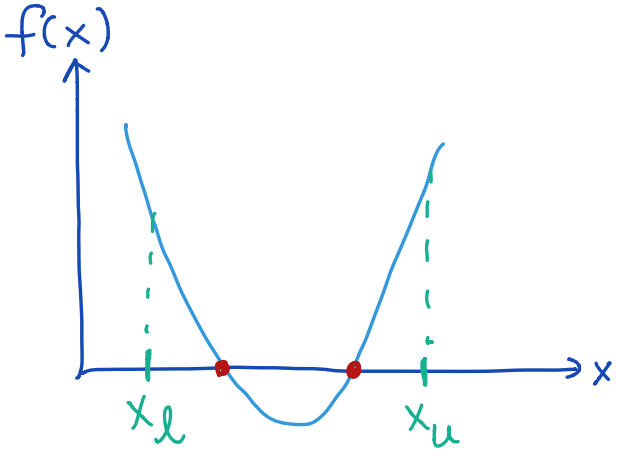


- Check the validity of the root by substituting it into the equation.
- Graphical methods are not precise
- Can be used as starting pts. of other numerical methods.



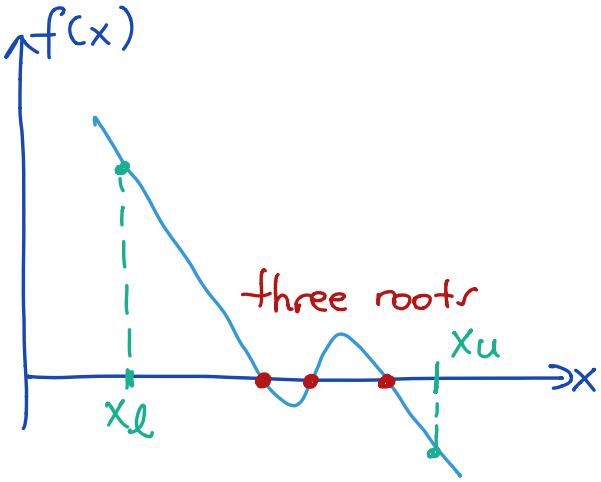
no roots





even # of roots between $[x_l, x_u]$

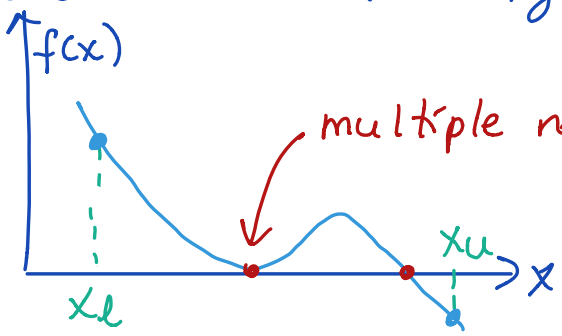
$f(x_l)$ and $f(x_u)$ have the same sign



$f(x_l)$ and $f(x_u)$ have opposite signs
$\Rightarrow$ odd number of roots in $[x_l, x_u]$

Exceptions

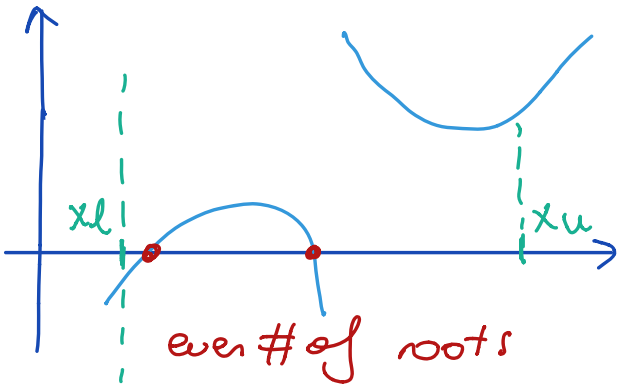
- functions that are tangential to x-axis.



e.g. $f(x) = (x-2)^2(x-4)$

multiple root at $x = 2$

- Discontinuous functions



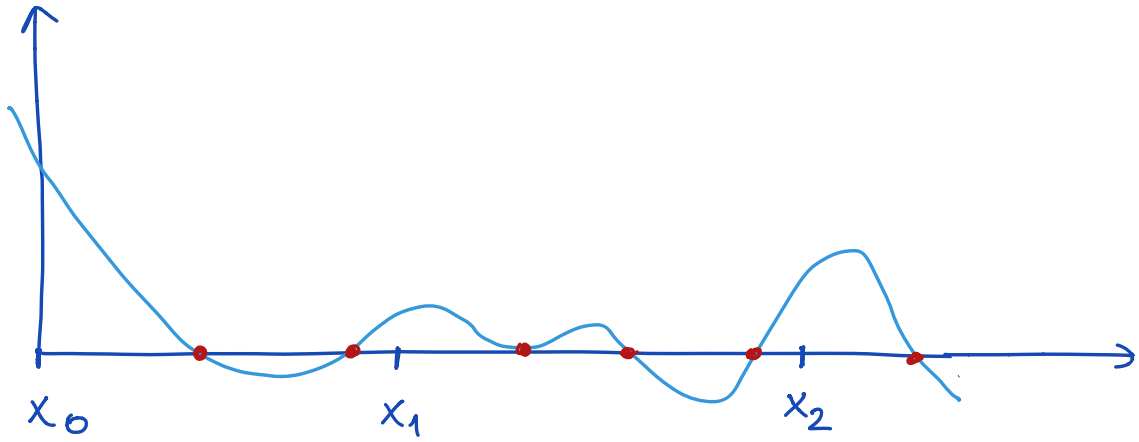
The Bisection Method

if $f(x_l)$ and $f(x_u)$ have opposite signs

$$f(x_l)f(x_u) < 0$$

then there is at least one real root between x_l and x_u.

⇒ How do we find x_l and x_u?



$|x_1 - x_0|$ too large ⇒ use smaller increments

- Make function evaluations at small increments across the region.
- When the function changes sign ⇒ assume there is a root within the increment.

- If length is too small ⇒ seach is too time consuming.

  if length is too large ⇒ miss closely spaced roots

Bisection Method (a.k.a binay chopping interval halving, Boltano's method)

Step 1: Choose x_l and x_u s.t. $f(x_l)f(x_u) < 0$

Step 2: Estimate root as:

$$x_r = \frac{x_l + x_u}{2}$$

Step 3: If $f(x_l)f(x_r) < 0$, root lies in lower subinterval

set $x_u = x_r$ return to step 2.

If $f(x_l)f(x_r) > 0$, root lies in upper subinterval

set $x_l = x_r$, return to step 2

If $f(x_l)f(x_r) = 0$

root $= x_r$, terminate

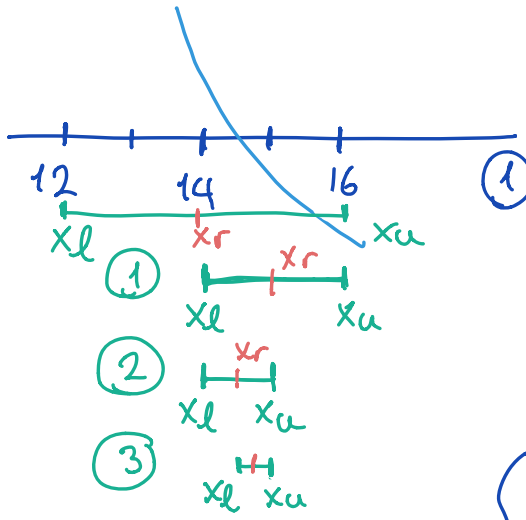
(or// if $\varepsilon_a = \left| \frac{x_r^{new} - x_r^{old}}{x_r^{new}} \right| \times 100\%$

if $\varepsilon_a < \varepsilon_{pr}$ stop)

⇒ **Idea**: The interval is always divided in half

Root is estimated at the midpoint.

EX True Root = 14.8011



① $x_l = 12$, $x_u = 16$

$x_r = \frac{12+16}{2} = 14$

$\varepsilon_t = 5.3\%$ (true percent relative error)

$f(x_l) f(x_r) > 0$

$\Rightarrow x_l = x_r = 14$

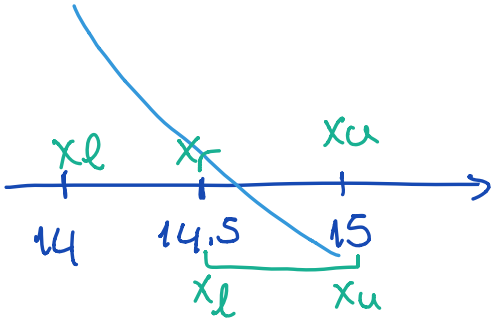
② $x_l = 14$, $x_u = 16$

$x_r = \frac{14+16}{2} = 15$

$\varepsilon_t = 1.3\%$

$f(x_l) f(x_r) = f(14) f(15) < 0$

$\Rightarrow x_u = x_r = 15$



③ $x_l = 14$, $x_u = 15$

$x_r = \frac{14+15}{2} = 14.5$ ($\varepsilon_t = 2\%$)

$f(x_l) f(x_r) = f(14) f(14.5) > 0$

$x_l = 14.5$, $x_u = 15$

- Notice that error does not decrease at each iteration.

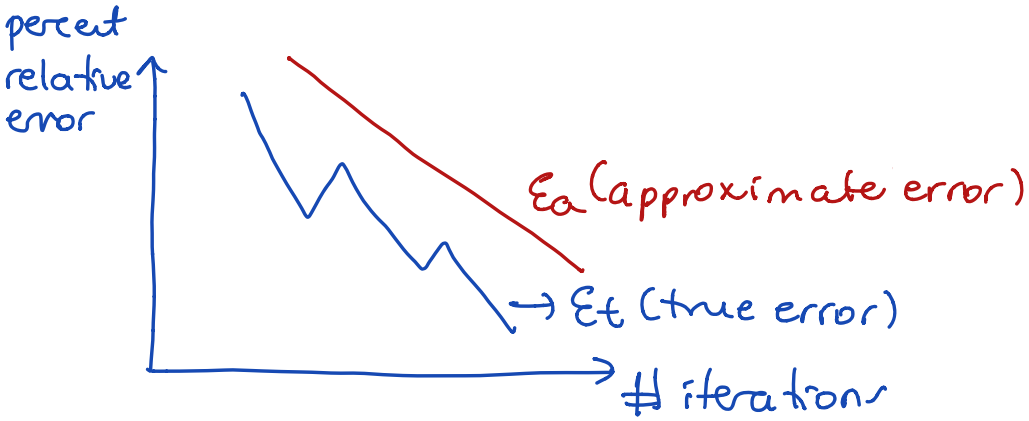
Approximate percent relative error

After 6 iterations $\varepsilon_s = 0.5\%$

$x_l = 14.75$

$x_u = 14.875$ $\varepsilon_a = 0.422\% < 0.5\%$

$x_r = 14.8125$



Since $\varepsilon_a > \varepsilon_t$, if $\varepsilon_a < \epsilon_s \Rightarrow \varepsilon_t < \epsilon_s$

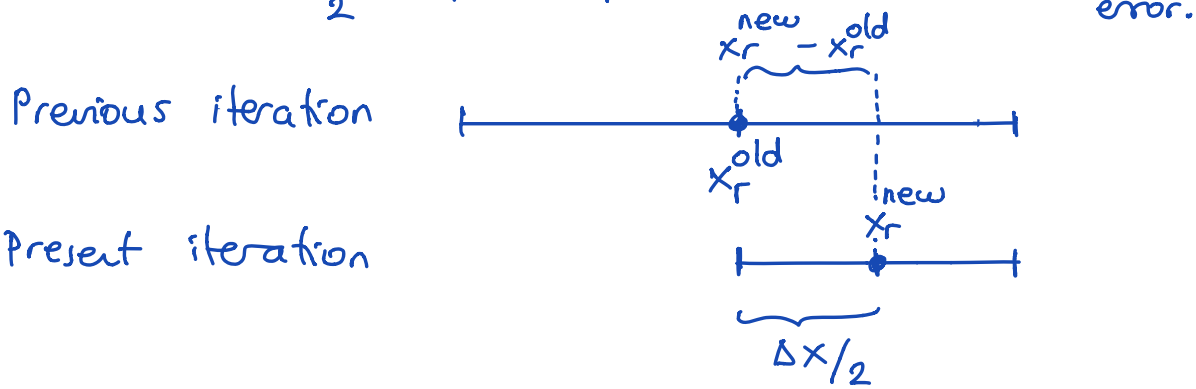
⊛ ε_a will always be greater than ε_t for the bisection method.

Since if $x_r = \dfrac{x_l + x_u}{2}$

we know the true root lies somewhere within an interval of $\dfrac{x_u - x_l}{2} = \dfrac{\Delta x}{2}$

Therefore, the root must lie somewhere within $\pm \dfrac{\Delta x}{2}$ of our estimate.

Because $\dfrac{\Delta x}{2} = x_r^{new} - x_r^{old}$ is an exact bound on true error.



The true root would never fall outside the bracketing interval.

⊛Another benefit of bisection method:

iterations required to attain an absolute error can be computed a priori (before starting the iterations)

iteration 0: $E_a^0 = x_u^0 - x_l^0 = \Delta x^0$

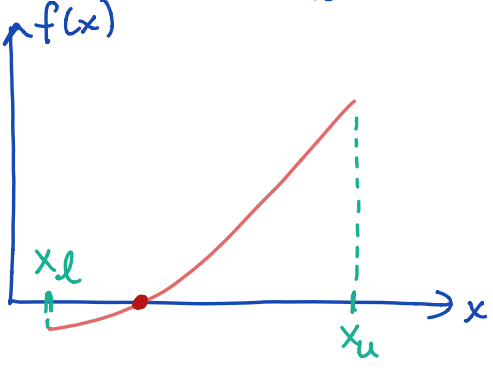
iteration 1: $E_a^1 = \frac{\Delta x^0}{2}$

iteration n: $E_a^n = \frac{\Delta x^0}{2^n} \Rightarrow n = \frac{\log\left(\frac{\Delta x^0}{E_{a,d}}\right)}{\log 2}$

$$n = \log_2\left(\frac{\Delta x^0}{E_{a,d}}\right)$$

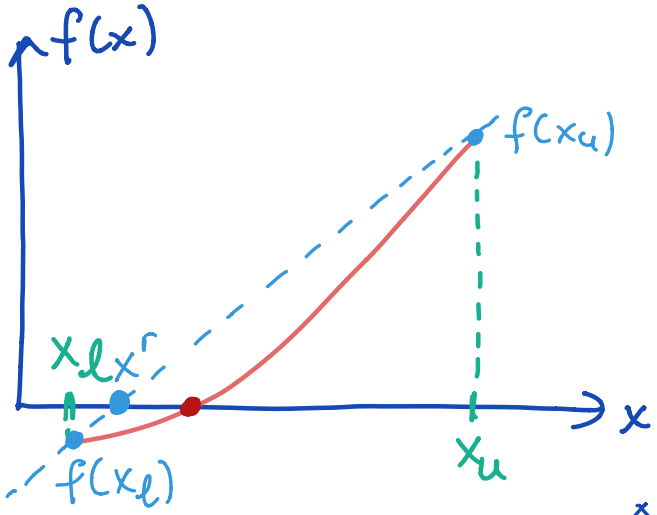
The False-Position Method

Shortcoming of bisection: in dividing interval $[x_l, x_u]$ into halves, no account is taken of the magnitude of $f(x_l)$ and $f(x_u)$. If $f(x_l) < f(x_u)$ it is likely that the root is closer to x_l than to x_u.



root is closer to x_l

Solution: join $f(x_l)$ and $f(x_u)$ by a straight line. The intersection of this line with the x axis gives an improved estimate of the root.



Method of false position
(Regula falsi)
or linear interpolation

Using similar triangles $\triangle x_l x_r f(x_l) \sim \triangle x_r x_u f(x_u)$

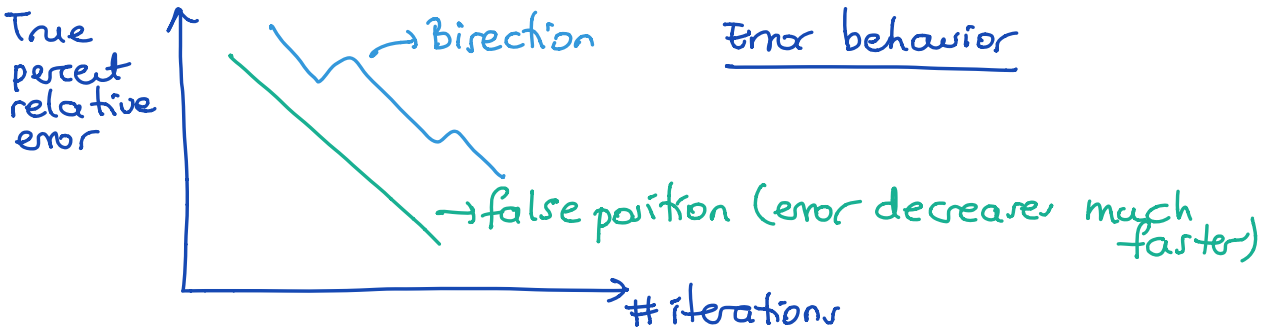
$$\frac{f(x_l)}{x_r - x_l} = \frac{f(x_u)}{x_r - x_u}$$

$$x_r = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$$ False-position formula

⇒ The value x_r then replaces x_l or x_u, whichever yields a function value with the same sign as $f(x_r)$.

⇒ The values x_l and x_u always bracket the root.

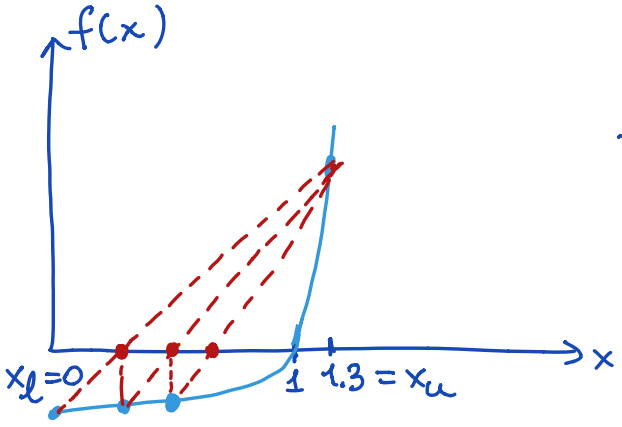
⇒ Repeat the process until error ε_a (approximate relative error) is small.



Error behavior

EX A case when bisection is preferable to False Position.

$f(x) = x^{10} - 1$ between $x = 0$ and $x = 1.3$



slow convergence of false position method.

iteration: 5

bisection $\varepsilon_t = 1.6\%$

$\varepsilon_a = 4.0\%$

false position $\varepsilon_t = 59.2\%$!!

$\varepsilon_a = 17.1\%$

Major weakness: one-side, i.e. one of the bracketing points stay fixed. ⇒ poor convergence